

Exact Cover Matrix

This table connects all possible candidates (on the left) with the constraints they correspond to (on the right). All 9x9x9=729 candidate possibilities are listed along with all 9x9x4=324 constraints. An actual exact cover matrix is a binary matrix -- all 1s and 0s. This version uses digits 1-9 simply for easier reading. Because we have some clues, some of these candidates are no longer possible -- they have no corresponding constraints. The exact cover problem amounts to finding the set of candidate-rows that "exactly covers" the columns with one value per column. The "cand/N" line counts the number of marks in each column. When this line reads all 1s, we are done. (If you solve the puzzle, you will see that this is the case.)

Sudoku Assistant, as all solvers do, solves this problem by investigating what the logical consequences are of selecting various subsets of candidate-rows. A naked pair -- four specific candidate-rows--for example, might require that certain columns have a mark in them. That would eliminate any candidates from the solution that would require additional marks there. The essence of the problem is to eliminate candidate-rows until just the "right" rows remain. To see how this works, check out these [exact cover examples](#).

3D Medusa, as all techniques, follows these connections to relatively simple logical inconsistencies. Hypothesis and (dis)proof goes further, looking for inconsistencies deeper along the solution.

An alternative, more efficient algorithm, the Dancing Links Algorithm (DLX), uses a clever and very efficient computational process of reversibly doubly-linked data structures to explore all possible solutions of a puzzle through brute-force trial and error. DLX is the method of choice for rapid finding of all possible solutions to Sudoku puzzles.

[illegible]

r1c6#9	9		9
r1c7#1	1	1	
r1c7#2	2	2	
r1c7#3	3	3	
r1c7#4	4	4	
r1c7#5	5	5	
r1c7#6	6	6	
r1c7#7	7	7	
r1c7#8	8	8	
r1c7#9	9	9	
r1c8#1	1	1	
r1c8#2	2	2	
r1c8#3	3	3	
r1c8#4	4	4	
r1c8#5	5	5	
r1c8#6	6	6	
r1c8#7	7	7	
r1c8#8	8	8	
r1c8#9	9	9	
r1c9#1	1	1	
r1c9#2	2	2	
r1c9#3	3	3	
r1c9#4	4	4	
r1c9#5	5	5	
r1c9#6	6	6	
r1c9#7	7	7	
r1c9#8	8	8	
r1c9#9	9	9	
r2c1#1	1		1
r2c1#2	2		2
r2c1#3	3		3
r2c1#4	4		4
r2c1#5	5		5
r2c1#6	6		6
r2c1#7	7		7
r2c1#8	8		8
r2c1#9	9		9
r2c2#1	1		1
r2c2#2	2		2
r2c2#3	3		3
r2c2#4	4		4
r2c2#5	5		5
r2c2#6	6		6
r2c2#7	7		7
r2c2#8	8		8
r2c2#9	9		9
r2c3#1	1		1
r2c3#2	2		2
r2c3#3	3		3
r2c3#4	4		4
r2c3#5	5		5
r2c3#6	6		6
r2c3#7	7		7
r2c3#8	8		8
r2c3#9	9		9
r2c4#1	1		1
r2c4#2	2		2
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r2c4#4	4		4
r2c4#5	5		5
r2c4#6	6		6
r2c4#7	7		7
r2c4#8	8		8
r2c4#9	9		9
r2c5#1	1		1
r2c5#2	2		2
r2c5#3	3		3
r2c5#4	4		4
r2c5#5	5		5
r2c5#6	6		6
r2c5#7	7		7
r2c5#8	8		8
r2c5#9	9		9
r2c6#1	1		1
r2c6#2	2		2
r2c6#3	3		3
r2c6#4	4		4
r2c6#5	5		5
r2c6#6	6		6
r2c6#7	7		7
r2c6#8	8		8
r2c6#9	9		9
r2c7#1	1		1
r2c7#2	2		2
r2c7#3	3		3
r2c7#4	4		4
r2c7#5	5		5
r2c7#6	6		6

r2c7#7	7	7
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r2c7#9	9	9
r2c8#1	1	1
r2c8#2	2	2
r2c8#3	3	3
r2c8#4	4	4
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r2c8#6	6	6
r2c8#7	7	7
r2c8#8	8	8
r2c8#9	9	9
r2c9#1	1	1
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r2c9#3	3	3
r2c9#4	4	4
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r2c9#6	6	6
r2c9#7	7	7
r2c9#8	8	8
r2c9#9	9	9
r3c1#1	1	1
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r3c1#4	4	4
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r3c1#7	7	7
r3c1#8	8	8
r3c1#9	9	9
r3c2#1	1	1
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r3c2#3	3	3
r3c2#4	4	4
r3c2#5	5	5
r3c2#6	6	6
r3c2#7	7	7
r3c2#8	8	8
r3c2#9	9	9
r3c3#1	1	1
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r3c3#3	3	3
r3c3#4	4	4
r3c3#5	5	5
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r3c4#9	9	9
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r3c5#4	4	4
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r3c5#6	6	6
r3c5#7	7	7
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r3c5#9	9	9
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r3c6#7	7	7
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r3c8#4	4	4

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r4c2#9	9	9
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r4c3#9	9	9
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r4c9#5	5		5
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r5c9#9	9		9

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r6c1#6	6		6
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r6c1#8	8		8
r6c1#9	9		9
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r6c6#9	9		9
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r6c7#6	6		6
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r6c7#9	9		9
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r6c8#9	9		9
r6c9#1	1		1
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r6c9#3	3		3
r6c9#4	4		4
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r7c1#6	6		6
r7c1#7	7		7
r7c1#8	8		8

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r7c2#2	2		2
r7c2#3	3		3
r7c2#4	4		4
r7c2#5	5		5
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r7c3#4	4		4
r7c3#5	5		5
r7c3#6	6		6
r7c3#7	7		7
r7c3#8	8		8
r7c3#9	9		9
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r7c4#4	4		4
r7c4#5	5		5
r7c4#6	6		6
r7c4#7	7		7
r7c4#8	8		8
r7c4#9	9		9
r7c5#1	1		1
r7c5#2	2		2
r7c5#3	3		3
r7c5#4	4		4
r7c5#5	5		5
r7c5#6	6		6
r7c5#7	7		7
r7c5#8	8		8
r7c5#9	9		9
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r7c6#3	3		3
r7c6#4	4		4
r7c6#5	5		5
r7c6#6	6		6
r7c6#7	7		7
r7c6#8	8		8
r7c6#9	9		9
r7c7#1	1		1
r7c7#2	2		2
r7c7#3	3		3
r7c7#4	4		4
r7c7#5	5		5
r7c7#6	6		6
r7c7#7	7		7
r7c7#8	8		8
r7c7#9	9		9
r7c8#1	1		1
r7c8#2	2		2
r7c8#3	3		3
r7c8#4	4		4
r7c8#5	5		5
r7c8#6	6		6
r7c8#7	7		7
r7c8#8	8		8
r7c8#9	9		9
r7c9#1	1		1
r7c9#2	2		2
r7c9#3	3		3
r7c9#4	4		4
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r7c9#6	6		6
r7c9#7	7		7
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r7c9#9	9		9
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r8c1#4	4		4
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r8c1#6	6		6
r8c1#7	7		7
r8c1#8	8		8
r8c1#9	9		9
r8c2#1	1		1
r8c2#2	2		2
r8c2#3	3		3
r8c2#4	4		4
r8c2#5	5		5
r8c2#6	6		6

r8c2#7	7		
r8c2#8	8		
r8c2#9	9		
r8c3#1	1		1
r8c3#2	2		2
r8c3#3	3		3
r8c3#4	4		
r8c3#5	5		
r8c3#6	6		
r8c3#7	7		
r8c3#8	8		
r8c3#9	9		
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r8c4#2	2		2
r8c4#3	3		3
r8c4#4	4		
r8c4#5	5		
r8c4#6	6		
r8c4#7	7		
r8c4#8	8		
r8c4#9	9		
r8c5#1	1		1
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r8c5#5	5		
r8c5#6	6		
r8c5#7	7		
r8c5#8	8		
r8c5#9	9		
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r8c6#2	2		2
r8c6#3	3		3
r8c6#4	4		
r8c6#5	5		
r8c6#6	6		
r8c6#7	7		
r8c6#8	8		
r8c6#9	9		
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r8c7#2	2		2
r8c7#3	3		3
r8c7#4	4		
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r8c7#6	6		
r8c7#7	7		
r8c7#8	8		
r8c7#9	9		
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r8c8#6	6		
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r8c8#8	8		
r8c8#9	9		
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r8c9#2	2		2
r8c9#3	3		3
r8c9#4	4		
r8c9#5	5		
r8c9#6	6		
r8c9#7	7		
r8c9#8	8		
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r9c2#8	8		
r9c2#9	9		
r9c3#1	1		
r9c3#2	2		
r9c3#3	3		
r9c3#4	4		

r9c3#5	5	
r9c3#6	6	
r9c3#7	7	
r9c3#8	8	
r9c3#9	9	

r9c4#1	1	
r9c4#2	2	
r9c4#3	3	
r9c4#4	4	
r9c4#5	5	
r9c4#6	6	
r9c4#7	7	
r9c4#8	8	
r9c4#9	9	

r9c5#1	1	
r9c5#2	2	
r9c5#3	3	
r9c5#4	4	
r9c5#5	5	
r9c5#6	6	
r9c5#7	7	
r9c5#8	8	
r9c5#9	9	

r9c6#1	1	
r9c6#2	2	
r9c6#3	3	
r9c6#4	4	
r9c6#5	5	
r9c6#6	6	
r9c6#7	7	
r9c6#8	8	
r9c6#9	9	

r9c7#1	1	
r9c7#2	2	
r9c7#3	3	
r9c7#4	4	
r9c7#5	5	
r9c7#6	6	
r9c7#7	7	
r9c7#8	8	
r9c7#9	9	

r9c8#1	1	
r9c8#2	2	
r9c8#3	3	
r9c8#4	4	
r9c8#5	5	
r9c8#6	6	
r9c8#7	7	
r9c8#8	8	
r9c8#9	9	

r9c9#1	1	
r9c9#2	2	
r9c9#3	3	
r9c9#4	4	
r9c9#5	5	
r9c9#6	6	
r9c9#7	7	
r9c9#8	8	
r9c9#9	9	
